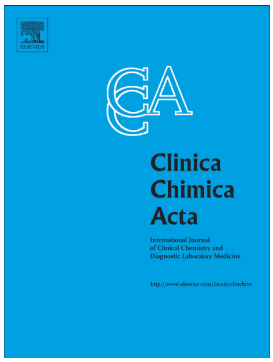
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miRNAs: Current and future diagnostic biomarkers in liver fibrosis

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miRNAs: current and future diagnostic biomarkers in liver fibrosis

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Abstract

Liver fibrosis is an abnormal response of the liver to various pathogenic factors. It represents a critical stage in the progression from chronic liver disease to cirrhosis and liver cancer. Without timely diagnosis and intervention, liver fibrosis can significantly reduce patients' survival rates and quality of life. In liver fibrosis, microRNA (miRNA) plays a dual role in disease progression by regulating gene expression and signaling pathways. Some miRNAs have antifibrotic effects by inhibiting the TGF-β/Smad signaling pathway or blocking the activation of hepatic stellate cells (HSCs). In contrast, other miRNAs accelerate fibrosis by activating the HIF-1α/VEGF pathway or promoting abnormal extracellular matrix deposition. This two-way relationship establishes a basis for investigating miRNAs as potential diagnostic markers. This narrative review synthesizes evidence on various miRNAs associated with liver fibrosis and their molecular mechanisms. It also evaluates their potential applications in serological testing. Research shows that miRNAs, because

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of their high stability and specificity, are likely to overcome the limitations of traditional methods for diagnosing liver fibrosis. Furthermore, this article evaluates the strengths and weaknesses of popular miRNA detection methods, such as real-time quantitative PCR, microarrays, and next-generation sequencing (NGS), emphasizing that optimizing standardized processes and high-throughput technologies is essential for facilitating the clinical use of miRNAs. In summary, miRNAs are promising non-invasive diagnostic tools for the early diagnosis and personalized treatment of liver fibrosis. Nonetheless, additional research is essential to improve the technology and clinical validation of miRNA detection for its widespread use in clinical settings.

Keywords

Liver fibrosis, miRNAs, miRNA detection methods, diagnostic markers.

Highlights

- Circulating miRNA profiles differ by liver-disease etiology, stage, and biological compartment.
- Pre-analytical control, normalization, and analytical validation are prerequisites for clinical translation.
- Multi-miRNA signatures combined with APRI or FIB-4 may improve non-invasive fibrosis classification.

1. Introduction

Liver fibrosis is an abnormal response to liver injury caused by various factors. It is characterized by the activation of hepatic stellate cells and the excessive deposition and abnormal distribution of extracellular matrix [1]. If not detected and treated promptly, liver fibrosis can progress to cirrhosis, liver cancer, or liver failure, significantly impacting patients' survival rates and quality of life [2]. Cirrhosis and liver cancer are estimated to rank 11th and 16th among global causes of death, respectively, resulting in approximately 1.16 million and 788,000 deaths each year [3]. Together with deaths from other liver diseases, around 2 million people die annually from liver disease, which accounts for 4% of total global mortality [4]. Liver fibrosis is a critical stage in the progression of liver diseases and holds potential for structural and functional recovery if diagnosed and treated promptly [5], thereby reducing the mortality rate caused by liver diseases.

Accurate detection of liver fibrosis allows for early diagnosis, enabling timely intervention. Clinical diagnosis of liver fibrosis currently relies on three main methods: liver tissue biopsy, imaging techniques, and serological tests. As the 'gold standard' for diagnosis, liver tissue biopsy is an invasive procedure that may lead to complications and

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sampling errors, which can negatively impact patient compliance [6]. The second method involves non-invasive imaging techniques, such as ultrasound liver elastography and magnetic resonance elastography [7]. However, the high equipment costs make examination fees expensive, making it unsuitable as the primary option for disease screening. Lastly, serological testing can identify biological markers associated with liver fibrosis, including hyaluronic acid and laminin. However, traditional serological testing has limitations. It cannot accurately determine the degree of fibrosis due to individual differences [8], and inflammation can elevate detection indicators, affecting the assessment of the condition [9]. Therefore, serological testing is typically used as an auxiliary tool for diagnosing liver fibrosis, providing indirect evidence for confirming the disease. Nonetheless, its advantages of being non-invasive and cost-effective are significant. It is possible to improve its diagnostic accuracy by discovering more specific and stable biological markers for liver fibrosis through serological testing, and combining its own advantages for clinical screening.

Research has shown that the expression of miRNAs in peripheral blood changes significantly during various diseases [10]. miRNA, a small RNA molecule, exhibits high specificity and stability in the progression of various diseases. miRNAs enter the bloodstream via exosomes and resist degradation by ribonucleases, allowing them to remain stable outside the cell. They regulate gene expression by binding specifically to their target genes, thereby influencing cellular functions [11]. They can be rapidly detected in various body fluids, including blood, saliva, and urine. Recently, researchers have identified miRNA's potential as diagnostic markers for diseases. This has resulted in numerous related studies [12]. To date, miRNA has been utilized as a crucial biomarker in the clinical diagnosis of certain diseases. For instance, abnormal expression of miR-21 indicates a potential risk for breast cancer [13]. Additionally, miR-18b is associated with predicting high-risk mantle cell lymphoma [14]. Similarly, miRNA also shows abnormal expression in liver diseases such as liver fibrosis [15], which provides a theoretical basis for applying miRNA to the diagnosis and condition assessment of liver fibrosis (Figure 1). This article will highlight several miRNAs with significant abnormal expression in liver fibrosis and their mechanisms of action. It will also explore the potential for these miRNAs to serve as future biomarkers for liver fibrosis, while considering the sensitivity, specificity, and cost-effectiveness of detection methods. Here, some methods for miRNA detection are also discussed.

Liver fibrosis develops through a wound-healing response that becomes persistent after chronic viral, metabolic, alcohol-related, cholestatic, or parasitic injury. Hepatocyte stress and cell death activate Kupffer cells and recruit inflammatory cells, which release transforming growth factor beta, platelet-derived growth factor, and other mediators. These signals convert quiescent hepatic stellate cells (HSCs) into contractile, extracellular-matrix-producing myofibroblasts. Progressive matrix deposition, matrix cross-linking, sinusoidal capillarization, and impaired matrix degradation increase portal resistance and distort hepatic architecture. miRNAs regulate this process at several levels: they control

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HSC activation and apoptosis, collagen synthesis, TGF-beta/SMAD signaling, macrophage polarization, and communication through extracellular vesicles [1,5,15,16]. The direction and magnitude of circulating changes may therefore differ from those in liver tissue and may vary among etiologies.

This is a narrative review of selected clinical and mechanistic studies on miRNAs in liver fibrosis. Human serum or plasma findings are considered alongside experimental studies of hepatic stellate-cell regulation and fibrogenesis. During revision, bibliographic records and selected study findings were checked against PubMed-indexed records accessed through Europe PMC and DOI-linked publication records. These checks support source verification; they do not constitute a systematic literature search. No protocol registration or quantitative meta-analysis is presented.

2. Introduction to miRNA

Mature miRNAs are small non-coding RNAs, usually 18 to 25 nucleotides long [17], with a typical length of approximately 22 nucleotides [18]. They regulate gene expression at the post-transcriptional level. The first miRNA, LIN-4, was discovered in Caenorhabditis elegans in 1993 [19]. Since then, researchers have identified about 30,424 miRNAs across 206 species [19]. In humans, there are approximately 700 distinct miRNAs [20]. Most miRNA genes are located within the introns of coding genes, although some are found in exons and untranslated regions (UTRs) [21]. The biogenesis of miRNAs starts when RNA polymerase II (Pol II) transcribes primary miRNAs (pri-miRNAs) [22]. These are subsequently processed by RNase III-type endonucleases to produce mature miRNA duplexes [18]. Mature miRNAs associate with Argonaute (AGO) proteins to form RNA-induced silencing complexes. These complexes regulate gene expression by base-pairing with the 3' or 5' untranslated regions of target mRNAs, thus performing their biological functions [21] (Figure 2). It is reported that miRNAs can control the expression of approximately 30% of essential human genes, which are crucial for survival and development [23]. Under physiological conditions, miRNAs regulate cellular proliferation, differentiation, apoptosis, organ development, and the maintenance of organ functions [24]. Typically, each miRNA can modulate the expression of multiple target mRNAs, and a single mRNA can be regulated by several miRNAs. In disease states, miRNA expression patterns change and closely correlate with disease progression [16]. In the liver, miRNAs play a crucial role in the onset and progression of liver diseases by regulating pathways like lipid metabolism and the activation or quiescence of hepatic stellate cells (HSCs) [15]. Studies have confirmed that abnormal miRNA expression is closely linked to liver diseases caused by viruses, parasites, and alcohol. Additionally, the levels of miRNAs in peripheral blood correlate with the onset and progression of liver diseases [25]. These findings provide a theoretical basis for using miRNAs as non-invasive diagnostic markers for liver fibrosis. They also encourage further research into the role of miRNAs in liver diseases. This article will discuss in detail the regulatory role of miRNAs in liver fibrosis and their potential clinical application prospects as potential biomarkers.

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3. Circulating miRNAs across liver-disease etiologies

Because viral hepatitis, metabolic dysfunction-associated steatotic liver disease, alcohol-associated liver disease, cholestatic disease, and schistosomiasis differ in inflammatory drivers and tissue injury, we organized the evidence by etiology whenever the primary studies allowed it. The miRNAs discussed below were selected because they recur across mechanistic studies and include human serum or plasma data; this is not an exhaustive catalogue. Other candidates, including miR-192, miR-194, miR-199a-5p, miR-221/222, miR-223, and liver-specific extracellular-vesicle miRNAs, are addressed in the broader literature but were outside the focused space of this review.

3.1 miR-122

miR-122 is a liver-specific miRNA that is highly abundant in hepatic tissue [26]. Under normal physiological conditions, miR-122 is essential for liver development, differentiation, homeostasis, and overall function [27]. In humans, miR-122 constitutes 52% of the total miRNA content in the liver, and its proportion is even higher in mice [28,29]. However, in pathological conditions, the expression of miR-122 alters, and it plays a role in disease regulation through various mechanisms. In early studies of miR-122, it was investigated as a tumor suppressor affecting the impact on hepatocellular carcinoma. Abnormal miR-122 expression is positively associated with poor prognosis and metastasis. It regulates the expression of oncogenes and tumor suppressor genes, as well as influences cell proliferation, apoptosis, and cancer cell metastasis [30].

As research advances, specific mechanisms affecting liver diseases have been identified. Studies indicate that miR-122 plays a role in chronic liver inflammation and lipid metabolism. For instance, alcohol reduces miR-122 expression in human and animal livers. This reduction promotes hypoxia-inducible factor-1α and peroxisome proliferator-activated receptor γ, resulting in steatosis and inflammation, which can lead to the progression of alcoholic liver disease into liver fibrosis. At this stage, restoring miR-122 levels can inhibit these pathways and improve alcoholic liver disease [31]. Researchers found that knocking out miR-122 in mice accelerated glutamine breakdown in the liver. Glutamine is a conditionally essential amino acid. Enhanced breakdown of glutamine promotes the activation of hepatic stellate cells, which leads to liver fibrosis [32]. Therefore, it is concluded that miR-122 can influence the progression of liver disease by regulating glutamine breakdown [33]. Researchers have conducted various experiments to explore the connection between miR-122 expression and liver fibrosis. In a liver fibrosis mouse model induced by carbon tetrachloride (CCl4), miR-122 levels were significantly reduced in HSCs. Experiments showed that miR-122 can target and bind to the 3' end of P4HA1, inhibiting its expression and reducing HSC activation. Overexpression of miR-122 inhibits the proliferation of the human hepatic stellate cell line (LX2) and slows liver fibrosis progression [34].

In a clinical analysis of 40 patients with chronic hepatitis C and 20 healthy individuals, serum miR-122 was higher in the patient group [35]. Although the reported AUC was

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1.00, the small single-centre sample and case-control design can overestimate discrimination, and miR-122 did not reliably distinguish fibrosis severity. Serum miR-122 should therefore be viewed as a candidate marker requiring prospective, etiology-stratified validation rather than as a stand-alone staging test.

3.2 miR-29: a key regulator in liver fibrosis

The human miR-29 family consists of three members: miR-29a, miR-29b, and miR-29c. Abnormal levels of miR-29 have been observed in various diseases, including osteoporosis, muscle atrophy, and diabetes [36,37,38]. Studies have shown that abnormal miR-29 levels are associated with the severity of fibrotic diseases [39]. miR-29 affects fibrosis progression through various pathways, including the TGF-β signaling pathway. TGF-β1 is a pivotal profibrotic cytokine that regulates the proliferation of fibrous connective tissue and promotes the synthesis and deposition of collagen, playing a crucial role in the development of fibrosis in nearly all organs [40]. Targeting signaling molecules from both the standard and alternative TGF-β pathways, such as Smad3 and Smad7, may be beneficial for treating fibrotic diseases [41].

In the liver, miR-29 has a clear protective role in preventing liver fibrosis. In a mouse model of liver fibrosis caused by a high-fat diet (HFD), miR-29a reduces the accumulation of subcutaneous, visceral, and intestinal fat. It also decreases hepatocellular steatosis and alleviates liver fibrosis by inhibiting CD36 [42]. During the onset of liver fibrosis, platelet-derived growth factor (PDGF) significantly stimulates the secretion of insulin-like growth factor-1 (IGF-1) and IGF-1 binding protein by rat HSCs. Both IGF-1 and PDGF stimulate DNA synthesis and promote the division and proliferation of rat HSCs [43]. High levels of miR-29 in HSCs inhibit the expression of IGF-1 and PDGF-C, thereby regulating the proliferation and differentiation of these cells [44]. In a mouse model of cholestatic liver injury and fibrosis caused by bile duct ligation (BDL), exogenous miR-29 was found to target and inhibit the expression of phosphoinositide 3-kinase p85α (PI3Kp85α), preventing the progression of liver fibrosis [45].

Research indicates that during the development of liver fibrosis, the expression level of miR-29 in the liver decreases, while the expression level in serum specimens increases. For example, in vitro isolation and culture of primary mouse HSCs show that the expression of miR-29 is reduced in activated HSCs. This reduction is due to the stimulation of HSCs by TGF-β or PDGF-BB, leading to autophagy that packages miR-29 into extracellular vesicles and expels them from HSCs. Consequently, the supernatant in the culture medium contains increased levels of miR-29, and liver fibrosis worsens further [46]. In a liver fibrosis mouse model treated with CCl4, the expression of miR-29 in peripheral blood is upregulated more than fivefold. In a study of serum samples from 44 patients with CHC and 12 healthy volunteers, the level of serum miR-29 in CHC patients was significantly increased (P=0.004), indicating that the change in miR-29 levels is associated with the occurrence of fibrotic diseases [46]. Due to the significant changes in miR-29 content in serum during the development of liver fibrosis and its high detection sensitivity, miR-29 has the potential to serve as a non-invasive diagnostic biomarker for liver

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fibrosis.

3.3 miR-34: a multifaceted regulator in liver fibrosis

The human miR-34 family includes three members: miR-34a, miR-34b, and miR-34c [47]. miR-34 gained attention because it is regulated by the tumor suppressor p53 and is expressed in various cancers [48]. It inhibits tumorigenesis and progression by promoting apoptosis, suppressing invasion and metastasis, reducing proliferation, enhancing differentiation, and boosting antitumor immune responses [49]. Furthermore, miR-34 also plays a role in regulating cellular senescence and stress responses [50,51]. The dysregulation of miR-34 expression is closely associated with the progression of fibrotic diseases, such as in aged pulmonary fibrosis where increased miR-34 expression exacerbates disease progression [52], while miR-34 knockout mice exhibit more severe fibrosis compared to wild-type mice in lung fibrosis injury models [53]. However, further investigation is needed to determine if miR-34 acts as a pro-fibrotic miRNA.

In the liver, miR-34a influences hepatocyte behavior during development and progression by encouraging cellular aging and cell death [54]. It affects various pathways related to liver fibrosis. For instance, it influences the miR-34a/Sirt1/p53 signaling pathway, which in turn inhibits the TGF-β1/Smads signaling pathway and reduces the development of liver fibrosis [55]. In a mouse model of liver injury caused by BDL, removing miR-34a specifically decreased aging and fibrosis responses in bile duct cells [56]. In a mouse model of alcohol-related liver disease (ALD), the levels of miR-34 increased. When miR-34 was knocked out in mice fed ethanol, the indicators of fibrosis and inflammation decreased. It was also observed that the markers of M2-type macrophages increased, while those of M1-type macrophages decreased [57]. Because macrophage effects are context-dependent and can vary with disease stage and tissue environment [58], miR-34 likely influences liver fibrosis by regulating the polarization of macrophages. Additionally, miR-34a binds to the 3' end of ACSL1, resulting in the deposition of extracellular matrix (ECM) proteins like collagen type I (Col1) and α-smooth muscle actin (α-SMA) in rats, and it promotes the activation of HSCs [59]. In summary, miR-34a exhibits profibrotic effects in the liver.

Clinical sample analysis has shown that serum miR-34 levels change with disease progression. A study involving 41 patients with CHC and 18 healthy individuals revealed that serum miR-34 levels correlate with the liver fibrosis index. Additionally, serum total bile acids (TBA), alanine aminotransferase (ALT), aspartate aminotransferase (AST), and the AST/ALT ratio also show positive correlations with miR-34. This suggests that miR-34 may serve as a marker for assessing fibrosis and prognosis in CHC patients [60]. Thus, miR-34 could serve as an additional indicator for disease staging and predicting treatment outcomes in clinical practice, offering new insights for diagnosing and treating liver fibrosis. (Table 1)

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4. Downregulated miRNAs in the serum of liver fibrosis patients

4.1 *let-7*: a critical modulator with reduced expression in liver fibrosis

let-7 is one of the first miRNAs to be discovered, and its naming convention has remained consistent over time [75]. The human genome encompasses 13 members of the *let-7* family [76]. Dysregulation of *let-7* is commonly linked to decreased cellular differentiation, which can lead to diseases like cancer [77]. *let-7* is closely associated with cancer, and its abnormal expression is considered an important biomarker for detecting various cancers in blood tests [78]. Elevated levels of specific *let-7* subtypes have been found in the plasma of patients with breast, prostate, and liver cancers, as well as in the urine of those with renal cell carcinoma. The notable changes in *let-7* levels in body fluids, combined with the simplicity of sample collection, provide opportunities for non-invasive disease detection [78].

let-7 expression in the liver is influenced by liver diseases, which affects disease progression [79]. Studies indicate that *let-7* inhibits apoptosis-related factors like FAS, reducing hepatocyte apoptosis. It also suppresses TGF-β1 production, which helps resist and prevent liver fibrosis [62]. In a mouse model of liver injury, expression levels of the *let-7* family were significantly reduced. *let-7* was found to inhibit activation markers of HSCs, such as Col1α1 and fibronectin 1 (FN1), thereby reducing HSC activation and alleviating liver fibrosis [63]. Upregulating *let-7* specifically in the liver can reduce liver fibrosis. Research also shows that miRNAs from other species, such as plants and parasites, can regulate host genes and can be detected within the host organism [80]. Based on these findings, researchers discovered that *sja-let-7*, derived from the extracellular vesicles of Schistosoma japonicum, can target and inhibit mouse Col1α2. It also downregulates the TGF-β/Smad signaling pathway, reducing HSC activation [79].

Changes in serum *let-7* family levels are closely linked to disease progression. A long-term follow-up study of 130 patients with CHC found that various miRNAs show abnormal expression as the disease progresses. The strongest correlation was observed between *let-7* levels and CHC liver fibrosis. Compared to 16 healthy volunteers, plasma levels of the evaluated *let-7* family members in CHC patients were reduced, and this reduction was negatively correlated with the progression of fibrosis [64]. Another study involving 163 patients with schistosomiasis-induced liver fibrosis, assessed by ultrasound for staging, revealed that the serum levels of *let-7a-5p* and *let-7d-5p* were negatively correlated with fibrosis staging [81]. The expression of *let-7a-5p* and *let-7d-5p* in patients with mild fibrosis (stages 0 and 1) was significantly higher than in those with moderate or severe fibrosis (stages 2 and 3) [81]. Considering the significant regulatory effect of *let-7* on liver fibrosis and its clear correlation with disease progression, incorporating *let-7* as a serological marker for non-invasive liver fibrosis diagnosis could be beneficial for future disease staging and monitoring.

4.2 miR-21: a biomarker with heterogeneous expression in liver fibrosis

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miR-21 is among the earliest discovered and most conserved miRNAs [82]. It plays a role in disease progression by targeting specific genes [83,84]. Because of its high expression levels and widespread presence in cells, changes in miR-21 are especially evident during disease onset [85]. miR-21 is now recognized as a crucial biomarker, as its abnormal expression is linked to the progression of many diseases [85]. For example, in patients with acute coronary syndrome (ACS), serum levels of miR-21 can increase to twice their normal levels [86]. In digestive system cancers, miR-21 has been identified as a biomarker with high diagnostic accuracy [87], providing a relatively reliable indicator for disease diagnosis.

In the liver, miR-21 influences liver fibrosis progression by mediating signaling pathways, and its serum levels are linked to disease advancement. In a mouse model of liver injury and fibrosis caused by arsenite, levels of both miR-21 and hypoxia-inducible factor 1 (HIF-1) are significantly increased. miR-21 can activate the HIF-1α/VEGF signaling pathway, facilitating the activation of HSCs and the development of liver fibrosis [65]. In contrast, miR-21 knockout reduces M2 macrophage polarization and attenuates arsenite-induced liver fibrosis in mice [66]. Furthermore, miR-21 plays a role in regulating the TGF-β signaling pathway, impacting liver fibrosis progression. Smad7 is a key modulator of liver fibrosis, acting as an inhibitor of the TGF-β1/Smad pathway [88]. miR-21 targets Smad7, affecting the TGF-β1/Smad pathway's role in HSC activation by regulating Smad7 expression [67]. In LX2 cells, polyphenols increase Smad7 expression by inhibiting miR-21, which helps to block the TGF-β1/Smad signaling pathway and reduce liver fibrosis [89].

An additional analysis explored serum miR-21 across fibrosis stages in 123 patients with hepatitis B virus infection. Serum miR-21 decreased from stages S0-S2 to S3-S4 ($p<0.05$) [68], whereas patients with autoimmune hepatitis had higher levels than healthy controls and patients with chronic hepatitis C [90]. These opposing directions illustrate etiology- and stage-dependent regulation. Overall, miR-21 levels in plasma vary substantially with liver disease and fibrosis stage; because inflammation, cancer, and other organ injury also alter miR-21, its specificity for liver fibrosis is limited. It may be useful only as part of a disease-specific panel.

4.3 miR-23/27/24: a cluster of miRNAs with altered expression in liver fibrosis

Some miRNA genes are located next to each other in the genome, forming what are called miRNA clusters. Currently, it is known that there are 159 miRNA clusters in the human genome [91]. A miRNA cluster consists of two or more miRNA-encoding genes that usually share similar expression mechanisms [92]. The members of a miRNA cluster may share the same targets or may target different genes within a particular pathway. In the human genome, the miR-23-27-24 cluster includes the miR-23a cluster (miR-23a-27a-24-2) located on chromosome 19 and the miR-23b cluster (miR-23b-27b-24-1) found on chromosome 9 [93]. Each cluster is transcribed as a single primary

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transcript, while the two clusters occupy distinct genomic loci and have separate regulatory regions. Research has shown that miR-23, miR-27, and miR-24 share overlapping targets; however, their biological functions are independent due to differences in how they are regulated [94]. Changes in their expression under disease conditions may lead to distinctly different regulatory effects on the disease [95].

In liver fibrosis, the miR-23/27/24 cluster of miRNAs plays different regulatory roles. miR-23b, for instance, can target and downregulate the expression of acyl-CoA thioesterase 4 (Acot4), inhibiting the degradation of acetyl-CoA and reducing free fatty acids, ultimately ameliorating liver fibrosis induced by non-alcoholic steatohepatitis [70]. Additionally, miR-23a binds to the tumor suppressor gene PTEN, resulting in decreased expression of E-cadherin and increased levels of α-SMA and Col1, which promotes liver fibrosis [69]. In a Schistosoma japonicum-induced liver fibrosis model, miR-27b regulates TGF-β1 mRNA stability through KSRP. Knockdown of miR-27b or overexpression of KSRP attenuated hepatic fibrosis [72]. In a mouse model of cholestatic liver disease, increased expression of the MEN1 gene raises TGF-β1 levels, which contributes to liver fibrosis. miR-24 binds to the MEN1 gene and inhibits its expression, reducing liver fibrosis. Conversely, when miR-24 expression is suppressed, liver fibrosis exacerbates [73]. Overexpression of miR-23b/27b/24-1 in mice with liver fibrosis, achieved through plasmid transfection, alleviates the condition. This occurs because miR-23b/27b/24-1 effectively inhibits the expression of liver fibrosis-related genes, including TGF-β2, Gremlin1, LOX, ITGα2, and ITGα5, thus preventing the activation of HSCs [74].

Researchers have observed changes in the serum levels of miR-23, miR-27, and miR-24 in liver disease (Figures 3 and 4). In patients with CHC, the level of miR-23b in serum is negatively correlated with AST levels. Additionally, miR-27a is significantly downregulated in CHC patients, with a decrease of about 0.198-fold ($p < 0.001$). This indicates that miR-27a levels in peripheral blood change significantly during liver injury [71]. However, no studies have statistically examined the expression of miR-24 in the serum of patients with liver fibrosis. In summary, given the varying impacts of miR-23, miR-27, and miR-24 on liver disease and their fluctuating expression levels, further research is needed to determine if these miRNAs can effectively assess liver cell injury, HSC activation, and liver fibrosis (Table 1).

5. Detection, normalization, and analytical validation of circulating miRNAs

Circulating miRNA assays require control of sequence specificity, low copy number, and pre-analytical variation. RT-qPCR remains practical for targeted validation, whereas microarrays and small-RNA sequencing support broader discovery. For normalization, no single reference is universally valid: candidate endogenous controls should be validated in the same matrix, global-mean normalization is useful for sufficiently large panels, synthetic spike-in oligonucleotides monitor extraction and reverse-transcription recovery, volume-based normalization reports copies per input volume, and absolute quantification can be achieved with calibrated standards or digital PCR. Hemolysis, anticoagulant

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choice, platelet contamination, storage time, and freeze-thaw cycles should be recorded and controlled. The reporting framework follows the MIQE principles [96].

5.1 Real-time quantitative polymerase chain reaction (qPCR)

RT-qPCR is widely used for targeted quantification of miRNAs [97]. This technique relies on the polymerase chain reaction and uses fluorescent probes or labeled primers. It quantitatively analyzes miRNA expression levels by monitoring fluorescence signal changes in real time during the PCR amplification process. The detection process involves two steps: first, miRNA templates are reverse-transcribed into complementary DNA (cDNA), and second, the reverse-transcribed cDNA is quantitatively detected using real-time fluorescent PCR [97]. Because of its high stability, operability, and the commercial availability of reagents, RT-qPCR is widely used for targeted miRNA quantification.

However, qPCR encounters challenges in miRNA detection, such as difficulties in primer design due to the short sequences of miRNAs and the risk of non-specific reactions during reverse transcription and PCR amplification [98]. To address these limitations, researchers use reverse-transcription qPCR and, where appropriate, rolling-circle amplification-based qPCR (RC-qPCR) to improve analytical specificity and sensitivity. Overall, qPCR is crucial for analyzing miRNA expression, particularly in the quantitative assessment of clinical samples and the examination of disease markers. Advances in technology and optimized methods are likely to make qPCR applications in miRNA detection more widespread and precise.

5.2 Microarray

Microarray is a high-throughput method for analyzing miRNA, allowing for the simultaneous quantification of multiple miRNAs in a single sample. It is commonly used to analyze differences in miRNA expression profiles. In this technique, a solid support is immobilized with a high-density array of known sequence probes. These probes can hybridize with specific miRNAs, enabling the analysis of miRNA expression profiles based on the intensity and distribution of the detected hybridization signals [99]. The high-throughput and quantitative analysis capabilities of microarray technology can complement the deficiencies of qPCR methods, enhancing the potential of traditional techniques [100]. Its main advantage is that it provides comprehensive information on miRNA expression, which helps in identifying expression patterns of disease-related miRNAs and potential biomarkers. Furthermore, microarray technology allows for the comparison of miRNA expression profiles under various conditions, offering crucial insights into the role of miRNAs in disease development.

However, microarray technology has several notable limitations. First, its high-throughput nature can lead to false-positive results. Additionally, improvements are needed in the cost of gene chip manufacturing, probe types, specificity, and result analysis, including probe design and data interpretation [101]. Microarray technology is widely used in

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research laboratories for preliminary screening and expression profiling of miRNA; however, its application in clinical settings still requires further validation and refinement. As technology advances, microarray technology is anticipated to become a crucial tool for understanding the role of miRNAs in diseases and offering new insights for diagnosis and treatment.

5.3 Next-generation sequencing (NGS)

In recent years, NGS technology has advanced significantly. Currently, NGS is used to analyze gene expression comprehensively through various methods, including transcriptome sequencing and miRNA sequencing. It can also detect miRNA expression profiles and other types of genetic material, such as DNA and RNA [102]. NGS enables both qualitative and quantitative detection of miRNA gene expression levels. This capability allows researchers to identify specific miRNA lengths and subtle sequence variations, making NGS an essential tool for precise miRNA detection [17]. NGS provides high-resolution and high-sensitivity miRNA expression data, enhancing our understanding of the complex roles of miRNAs in biological processes and diseases. Moreover, NGS can identify novel and unknown miRNAs, creating new opportunities for miRNA functional research and biomarker discovery.

Analytical validation should report limit of blank, limit of detection, limit of quantification, linearity, precision, amplification efficiency, and potential cross-reactivity with closely related miRNAs. Batch effects and run-to-run variation should be assessed with interspersed controls and replicated samples, and inter-laboratory comparability requires common calibrators, harmonized extraction protocols, and transparent reporting. Droplet digital PCR can quantify low-abundance miRNAs without a standard curve; CRISPR-associated nucleases and isothermal amplification formats are emerging options for rapid, instrument-light or point-of-care testing. These platforms remain promising but require orthogonal confirmation and clinical validation before routine use; ddPCR absolute-quantification principles are described in [103].

Method selection should match the clinical question, specimen type, required sensitivity, turnaround time, and available quality controls. A short targeted panel may be preferable for clinical validation, whereas sequencing is most useful for discovery. Newer digital, CRISPR-based, and isothermal assays should be evaluated against RT-qPCR with the same specimens and predefined analytical acceptance criteria. Table 2 summarizes the established measurement platforms.

6. Prospects and challenges of miRNA as diagnostic biomarkers

Diagnostic biomarkers are objective indicators that assess the presence of various diseases. Technological advancements have made these indicators more precise and reliable. Blood-based diagnostic biomarkers typically include proteins like ALT and AST for liver disease diagnosis, as well as carcinoembryonic antigen for various cancers [104,105]. To

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advance personalized medicine, there is an urgent need to discover new and more accurate biomarkers. Ideal diagnostic biomarkers should have characteristics including ease of sample collection, specificity to pathological changes, high sensitivity, and the ability to reflect disease severity or treatment progress [106]. As research on miRNAs advances, their recognized characteristics show potential as diagnostic biomarkers.

Many studies have found that certain miRNAs show altered expression in plasma or serum as diseases progress. For instance, miRNA-122-5p has been consistently elevated in serum and plasma samples from hepatocellular carcinoma patients, aiding in their identification with high specificity and sensitivity [107]. Additionally, abnormal miRNA expression in diseases like multiple sclerosis makes them valuable clinical diagnostic indicators [108]. Research has confirmed that miRNAs are sensitive markers that change with disease progression or treatment response [109]. miRNAs provide a non-invasive method for clinical testing that accurately diagnoses diseases, predicts disease progression, guides treatment, and assesses treatment responsiveness. These benefits position miRNAs as ideal candidates for diagnostic biomarkers across various diseases.

A clinically useful strategy is likely to combine a multi-miRNA signature with established scores rather than replace them. APRI and FIB-4 use routine laboratory variables and have complementary strengths and weaknesses [110,111]. A prespecified model that adds a small, etiology-stratified miRNA panel to APRI or FIB-4 could improve discrimination at intermediate stages, but apparent gains must be tested in external cohorts, decision-curve analyses, and calibration studies. The model should be locked before validation to avoid optimism from data-driven feature selection.

Recent reviews emphasize that circulating liver-specific miRNAs and noncoding RNAs provide a broader candidate space [112,113]. We therefore clarified that the present review is selective and focused on recurrent mechanistic evidence plus human serum or plasma observations; omission does not imply that other miRNAs lack biological or diagnostic relevance.

Furthermore, ideal biomarkers must have easily obtainable samples, which apply to miRNAs that can be readily extracted from liquid biopsies in blood, urine, and other body fluids. They also show high specificity for different tissues or cell types and can change with disease progression, demonstrating sensitivity useful for disease staging studies [12]. Developing new miRNA detection kits using nucleic acid technology is faster and more cost-effective than creating new antibodies for protein-based disease biomarkers. miRNAs offer diagnostic advantages over traditional biomarkers by demonstrating higher tissue specificity and lower cross-reactivity, thus enhancing diagnostic accuracy. For instance, miR-122 exhibits high specificity for liver diseases and does not show abnormal expression in plasma from other organ pathologies, unlike ALT and AST, which can increase due to issues in other organs. When test results are analyzed alongside liver tissue histopathology, miR-122 demonstrates higher diagnostic sensitivity compared to ALT [114]. Clearly,

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miRNA can be used as a serological diagnostic biomarker. This approach may enhance diagnostic accuracy and allow for the screening of diseases with high specificity, indicating significant potential for application.

However, using miRNAs as diagnostic biomarkers presents several challenges. Firstly, the expression of miRNAs can be influenced by various factors, such as inflammation and cancer, which may reduce their reliability for specific diseases [85]. Because miR-21 levels in the blood are relatively nonspecific for various diseases, it is currently regarded as the "most non-specific biomarker" [85]. An effective serum biomarker for diseases should exhibit high specificity. Secondly, a certain level of repeatability is essential for the detection and verification of diseases. Errors in detection methods, analysis processes, and statistical methods can affect repeatability (i.e., consistency) [115]. This limitation significantly impacts the use of serum miRNAs as diagnostic tools. Before clinically applying new disease biomarkers, it is essential to establish standard operating procedures. These should include sample size, type of anticoagulant, exclusion of hemolysis errors, and guidelines for sample processing and storage to ensure consistent and reliable results [116]. Many serum miRNA biomarkers have been validated in various populations and disease conditions, yet they have not yet entered clinical application. The challenges include the use of different measurement platforms, the absence of unified standards and reference ranges, and factors such as race, age, gender, sample type, and processing methods. These issues contribute to significant inconsistency and non-reproducibility across studies, hindering clinical implementation.

Despite the challenges, researchers are actively pursuing the development of miRNA as diagnostic biomarkers. By addressing technical challenges, standardization, and clinical validation issues, miRNAs are expected to become important tools in the future diagnosis and management of diseases. Researchers need to explore the biological functions and clinical relevance of specific miRNAs, optimize detection technologies, and establish unified measurement standards and reference ranges to effectively apply miRNAs in clinical practice. With ongoing research and advancements in technology, miRNA-based diagnostic tools are anticipated to revolutionize personalized and precision medicine.

7. Summary and Discussion

The discovery of miRNAs is a significant breakthrough in modern science. It alters our understanding of cellular biology and medical science, offering new perspectives for disease diagnosis and treatment. miRNAs influence physiological processes and regulate the occurrence and development of diseases. Recently, miRNAs have gained attention as promising biomarkers due to their non-invasive nature, specificity to tissues and diseases, and high stability across various body fluids. They are actively secreted through extracellular vesicles or released into peripheral blood during processes like apoptosis and necrosis. Changes in miRNA levels in peripheral blood are significantly linked to disease

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progression. In the future, quantitative detection of specific miRNAs in serum may be used to screen for liver fibrosis and assess disease severity. This serological diagnostic approach could circumvent the drawbacks of pathological biopsy and imaging examinations.

This article examines recurrent miRNA changes reported in serum or plasma during liver fibrosis, while emphasizing that direction and magnitude vary by etiology, stage, specimen, and assay. miR-122, miR-29, and miR-34 have been reported as increased in selected cohorts, whereas *let-7* and members of the miR-23/27/24 cluster are decreased in other cohorts; miR-21 shows particularly strong heterogeneity. The mechanistic sections explain how these miRNAs influence HSC activation, inflammatory signaling, and extracellular-matrix turnover. Their clinical use will require standardized pre-analytics, validated normalization, analytical performance studies, and prospective external validation.

The text also describes three common methods for detecting miRNAs, each of which has methodological limitations and is undergoing improvements. Currently, no optimal strategy exists for analyzing miRNAs in serum. Variations in experimental methods and technical approaches can lead to differing results, creating significant debate over the reliability of miRNAs as biomarkers for liver fibrosis detection. To address these issues, a unified experimental protocol and standard operating procedures must be developed for assessing miRNAs in serum, including guidelines for selecting starting materials, separation methods, and standardized detection and analysis procedures. Ongoing research focuses on developing and optimizing new methods. Additionally, improving and standardizing existing technologies, along with discovering new measurement methods, could shift the diagnosis of liver fibrosis using miRNAs from the laboratory to clinical practice.

Finally, the future prospects and challenges of using miRNAs as diagnostic markers in clinical settings are discussed. While miRNAs show great promise for significant breakthroughs in non-invasive disease detection, the field is still in its infancy and has a long way to go to reach its goals. Researchers are dedicated to identifying miRNAs that are highly specific to diseases and understanding how they function. By improving detection technology, creating standardized procedures, and strengthening clinical validation, researchers can conduct enough experiments to confirm the changes in miRNAs as statistically significant diagnostic markers as the disease advances. It is also essential to clearly outline the limitations of the experiments and remind other researchers and clinical staff to follow the recommended protocols for research or diagnostic testing. If we continue to build on our current research momentum, we will soon enter an era of rapid and accurate disease detection using serum miRNA-based biomarkers.

Competing Interests statement

The authors declare no conflict of interest

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Author contributions

YJL: conceptualization & writing-original draft; MA: conceptualization & writing-original draft; HY: conceptualization & writing-review and editing; ZYW: Draw figures & writing-review; LHL: conceptualization & supervision & writing-review and editing; GFC: conceptualization & supervision & writing-review and editing. All authors read and approved the final manuscript.

References

[1] Kisseleva T, Brenner D. Molecular and cellular mechanisms of liver fibrosis and its regression. Nat Rev Gastroenterol Hepatol. 2021;18:151-166.doi;10.1038/s41575-020-00372-7

[2] Bataller R, Brenner DA. Liver fibrosis. J Clin Invest. 2005;115:209-218.doi;10.1172/jci24282

[3] Asrani SK, Devarbhavi H, Eaton J, Kamath PS. Burden of liver diseases in the world. J Hepatol. 2019;70:151-171.doi;10.1016/j.jhep.2018.09.014

[4] Devarbhavi H, Asrani SK, Arab JP, Nartey YA, Pose E, Kamath PS. Global burden of liver disease: 2023 update. J Hepatol. 2023;79:516-537.doi;10.1016/j.jhep.2023.03.017

[5] Roehlen N, Crouchet E, Baumert TF. Liver Fibrosis: Mechanistic Concepts and Therapeutic Perspectives. Cells. 2020; 9(4):875. doi;10.3390/cells9040875

[6] Regev A, Berho M, Jeffers LJ, Milikowski C, Molina EG, Pyrsopoulos NT, Feng ZZ, Reddy KR, Schiff ER. Sampling error and intraobserver variation in liver biopsy in patients with chronic HCV infection. Am J Gastroenterol. 2002;97:2614-2618.doi;10.1111/j.1572-0241.2002.06038.x

[7] Yu JH, Lee HA, Kim SU. Noninvasive imaging biomarkers for liver fibrosis in nonalcoholic fatty liver disease: current and future. Clin Mol Hepatol. 2023;29:S136-s149.doi;10.3350/cmh.2022.0436

[8] Rossi E, Adams LA, Ching HL, Bulsara M, MacQuillan GC, Jeffrey GP. High biological variation of serum hyaluronic acid and Hepascore, a biochemical marker model for the prediction of liver fibrosis. Clin Chem Lab Med. 2013;51:1107-1114.doi;10.1515/cclm-2012-0584

[9] Castera L, Hartmann DJ, Chapel F, Guettier C, Mall F, Lons T, Richardet JP, Grimbert S, Morassi O, Beaugrand M, Trinchet JC. Serum laminin and type IV collagen are

Journal Pre-proof

Journal Pre-proof

accurate markers of histologically severe alcoholic hepatitis in patients with cirrhosis. J Hepatol. 2000;32:412-418.doi;10.1016/s0168-8278(00)80391-8

[10] Soifer HS, Rossi JJ, Saetrom P. miRNAs in disease and potential therapeutic applications. Mol Ther. 2007;15:2070-2079.doi;10.1038/sj.mt.6300311

[11] Bhaskaran M, Mohan M. miRNAs: history, biogenesis, and their evolving role in animal development and disease. Vet Pathol. 2014;51:759-774.doi;10.1177/0300985813502820

[12] Mori MA, Ludwig RG, Garcia-Martin R, Brandão BB, Kahn CR. Extracellular miRNAs: From Biomarkers to Mediators of Physiology and Disease. Cell Metab. 2019;30:656-673.doi;10.1016/j.cmet.2019.07.011

[13] Bertoli G, Cava C, Castiglioni I. miRNAs: New Biomarkers for Diagnosis, Prognosis, Therapy Prediction and Therapeutic Tools for Breast Cancer. Theranostics. 2015;5:1122-1143.doi;10.7150/thno.11543

[14] Husby S, Ralfkiaer U, Garde C, Zandi R, Ek S, Kolstad A, Jerkeman M, Laurell A, Räty R, Pedersen LB, Pedersen A, Ehinger M, Sundström C, Karjalainen-Lindsberg ML, Delabie J, Clasen-Linde E, Brown P, Cowland JB, Workman CT, Geisler CH, Grønbæk K. miR-18b overexpression identifies mantle cell lymphoma patients with poor outcome and improves the MIPI-B prognosticator. Blood. 2015;125:2669-2677.doi;10.1182/blood-2014-06-584193

[15] Wang X, He Y, Mackowiak B, Gao B. miRNAs as regulators, biomarkers and therapeutic targets in liver diseases. Gut. 2021;70:784-795.doi;10.1136/gutjnl-2020-322526

[16] Huang J, Yu X, Fries JW, Zhang L, Odenthal M. miRNA function in the profibrogenic interplay upon chronic liver disease. Int J Mol Sci. 2014;15:9360-9371.doi;10.3390/ijms15069360

[17] Liu K, Tong H, Li T, Wang X, Chen Y. Research progress in molecular biology related quantitative methods of MicroRNA. Am J Transl Res. 2020;12:3198-3211

[18] Ha M, Kim VN. Regulation of microRNA biogenesis. Nat Rev Mol Cell Biol. 2014;15:509-524.doi;10.1038/nrm3838

[19] Kozomara A, Griffiths-Jones S. miRBase: annotating high confidence miRNAs using deep sequencing data. Nucleic Acids Res. 2014;42:D68-73.doi;10.1093/nar/gkt1181

[20] Griffiths-Jones S, Saini HK, van Dongen S, Enright AJ. miRBase: tools for miRNA genomics. Nucleic Acids Res. 2008;36:D154-158.doi;10.1093/nar/gkm952

[21] Krol J, Loedige I, Filipowicz W. The widespread regulation of miRNA biogenesis, function and decay. Nat Rev Genet. 2010;11:597-610.doi;10.1038/nrg2843

[22] Bartel DP. miRNAs: genomics, biogenesis, mechanism, and function. Cell. 2004;116:281-297.doi;10.1016/s0092-8674(04)00045-5

[23] Filipowicz W, Bhattacharyya SN, Sonenberg N. Mechanisms of post-transcriptional regulation by miRNAs: are the answers in sight? Nat Rev Genet.

Journal Pre-proof

Journal Pre-proof

2008;9:102-114.doi;10.1038/nrg2290

[24] Bandiera S, Pfeffer S, Baumert TF, Zeisel MB. miR-122--a key factor and therapeutic target in liver disease. J Hepatol. 2015;62:448-457.doi;10.1016/j.jhep.2014.10.004

[25] Li H, Liu T, Yang Y, Cho WC, Flynn RJ, Harandi MF, Song H, Luo X, Zheng Y. Interplays of liver fibrosis-associated miRNAs: Molecular mechanisms and implications in diagnosis and therapy. Genes Dis. 2023;10:1457-1469.doi;10.1016/j.gendis.2022.08.013

[26] Hsu SH, Wang B, Kota J, Yu J, Costinean S, Kutay H, Yu L, Bai S, La Perle K, Chivukula RR, Mao H, Wei M, Clark KR, Mendell JR, Caligiuri MA, Jacob ST, Mendell JT, Ghoshal K. Essential metabolic, anti-inflammatory, and anti-tumorigenic functions of miR-122 in liver. J Clin Invest. 2012;122:2871-2883.doi;10.1172/jci63539

[27] Hu J, Xu Y, Hao J, Wang S, Li C, Meng S. MiR-122 in hepatic function and liver diseases. Protein Cell. 2012;3:364-371.doi;10.1007/s13238-012-2036-3

[28] Girard M, Jacquemin E, Munnich A, Lyonnet S, Henrion-Caude A. miR-122, a paradigm for the role of miRNAs in the liver. J Hepatol. 2008;48:648-656.doi;10.1016/j.jhep.2008.01.019

[29] Lagos-Quintana M, Rauhut R, Yalcin A, Meyer J, Lendeckel W, Tuschl T. Identification of tissue-specific miRNAs from mouse. Curr Biol. 2002;12:735-739.doi;10.1016/s0960-9822(02)00809-6

[30] Al-Gazally ME, Khan R, Imran M, Ramírez-Coronel AA, Alshahrani SH, Altalbawy FMA, Turki Jalil A, Romero-Parra RM, Zabibah RS, Shahid Iqbal M, Karampoor S, Mirzaei R. The role and mechanism of action of miRNA-122 in cancer: Focusing on the liver. Int Immunopharmacol. 2023;123:110713.doi;10.1016/j.intimp.2023.110713

[31] Satishchandran A, Ambade A, Rao S, Hsueh YC, Iracheta-Vellve A, Tornai D, Lowe P, Gyongyosi B, Li J, Catalano D, Zhong L, Kodys K, Xie J, Bala S, Gao G, Szabo G. miRNA 122, Regulated by GRLH2, Protects Livers of Mice and Patients From Ethanol-Induced Liver Disease. Gastroenterology. 2018;154:238-252.e237.doi;10.1053/j.gastro.2017.09.022

[32] Yan R, Cai H, Zhou X, Bao G, Bai Z, Ge RL. Hypoxia-inducible factor-2α promotes fibrosis in non-alcoholic fatty liver disease by enhancing glutamine catabolism and inhibiting yes-associated protein phosphorylation in hepatic stellate cells. Front Endocrinol (Lausanne). 2024;15:1344971.doi;10.3389/fendo.2024.1344971

[33] Sengupta D, Cassel T, Teng KY, Aljuhani M, Chowdhary VK, Hu P, Zhang X, Fan TW, Ghoshal K. Regulation of hepatic glutamine metabolism by miR-122. Mol Metab. 2020;34:174-186.doi;10.1016/j.molmet.2020.01.003

[34] Li J, Ghazwani M, Zhang Y, Lu J, Li J, Fan J, Gandhi CR, Li S. miR-122 regulates collagen production via targeting hepatic stellate cells and suppressing P4HA1 expression.

Journal Pre-proof

J Hepatol. 2013;58:522-528.doi;10.1016/j.jhep.2012.11.011

[35] Omran AA, Osman KS, Kamel HM, Abdel-Naem EA, Hasan DE. miRNA-122 as a Novel Non-Invasive Marker of Liver Fibrosis in Hepatitis C Virus Patients. Clin Lab. 2016;62:1329-1337.doi;10.7754/Clin.Lab.2015.151141

[36] Lian WS, Ko JY, Chen YS, Ke HJ, Hsieh CK, Kuo CW, Wang SY, Huang BW, Tseng JG, Wang FS. miRNA-29a represses osteoclast formation and protects against osteoporosis by regulating PCAF-mediated RANKL and CXCL12. Cell Death Dis. 2019;10:705.doi;10.1038/s41419-019-1942-1

[37] Wang H, Wang B, Zhang A, Hassounah F, Seow Y, Wood M, Ma F, Klein JD, Price SR, Wang XH. Exosome-Mediated miR-29 Transfer Reduces Muscle Atrophy and Kidney Fibrosis in Mice. Mol Ther. 2019;27:571-583.doi;10.1016/j.ymthe.2019.01.008

[38] Dalgaard LT, Sørensen AE, Hardikar AA, Joglekar MV. The miRNA-29 family: role in metabolism and metabolic disease. Am J Physiol Cell Physiol. 2022;323:C367-c377.doi;10.1152/ajpcell.00051.2022

[39] Chioccioli M, Roy S, Newell R, Pestano L, Dickinson B, Rigby K, Herazo-Maya J, Jenkins G, Ian S, Saini G, Johnson SR, Braybrooke R, Yu G, Sauler M, Ahangari F, Ding S, DeIuliis J, Aurelien N, Montgomery RL, Kaminski N. A lung targeted miR-29 mimic as a therapy for pulmonary fibrosis. EBioMedicine. 2022;85:104304.doi;10.1016/j.ebiom.2022.104304

[40] Li L, Liu Y, Wang K, Mo J, Weng Z, Jiang H, Jin C. Stem cell exosomes: new hope and future potential for relieving liver fibrosis. Clin Mol Hepatol. 2024 Nov 7. doi; 10.3350/cmh.2024.0854

[41] Song Y, Wei J, Li R, Fu R, Han P, Wang H, Zhang G, Li S, Chen S, Liu Z, Zhao Y, Zhu C, Zhu J, Zhang S, Pei H, Cheng J, Wu J, Dong L, Song G, Shen X, Yao Q. Tyrosine kinase receptor B attenuates liver fibrosis by inhibiting TGF-β/SMAD signaling. Hepatology. 2023;78:1433-1447.doi;10.1097/hep.0000000000000319

[42] Lin HY, Wang FS, Yang YL, Huang YH. miRNA-29a Suppresses CD36 to Ameliorate High Fat Diet-Induced Steatohepatitis and Liver Fibrosis in Mice. Cells. 2019;8(10):1298.doi;10.3390/cells8101298

[43] Sobrevals L, Rodriguez C, Romero-Trevejo JL, Gondi G, Monreal I, Pañeda A, Juanarena N, Arcelus S, Razquin N, Guembe L, González-Aseguinolaza G, Prieto J, Fortes P. Insulin-like growth factor I gene transfer to cirrhotic liver induces fibrolysis and reduces fibrogenesis leading to cirrhosis reversion in rats. Hepatology. 2010;51:912-921.doi;10.1002/hep.23412

Journal Pre-proof

Journal Pre-proof

[44] Sekiya Y, Ogawa T, Yoshizato K, Ikeda K, Kawada N. Suppression of hepatic stellate cell activation by miRNA-29b. Biochem Biophys Res Commun. 2011;412:74-79.doi;10.1016/j.bbrc.2011.07.041

[45] Yang YL, Wang FS, Lin HY, Huang YH. Exogenous Therapeutics of Microrna-29a Attenuates Development of Hepatic Fibrosis in Cholestatic Animal Model through Regulation of Phosphoinositide 3-Kinase p85 Alpha. Int J Mol Sci. 2020; 21(10):3636. doi;10.3390/ijms21103636

[46] Yu X, Elfimova N, Müller M, Bachurski D, Koitzsch U, Drebber U, Mahabir E, Hansen HP, Friedman SL, Klein S, Dienes HP, Hösel M, Buettner R, Trebicka J, Kondylis V, Mannaerts I, Odenthal M. Autophagy-Related Activation of Hepatic Stellate Cells Reduces Cellular miR-29a by Promoting Its Vesicular Secretion. Cell Mol Gastroenterol Hepatol. 2022;13:1701-1716.doi;10.1016/j.jcmgh.2022.02.013

[47] He X, He L, Hannon GJ. The guardian's little helper: miRNAs in the p53 tumor suppressor network. Cancer Res. 2007;67:11099-11101.doi;10.1158/0008-5472.Can-07-2672

[48] Hermeking H. miRNAs in the p53 network: micromanagement of tumour suppression. Nat Rev Cancer. 2012;12:613-626.doi;10.1038/nrc3318

[49] Shi X, Kaller M, Rokavec M, Kirchner T, Horst D, Hermeking H. Characterization of a p53/miR-34a/CSF1R/STAT3 Feedback Loop in Colorectal Cancer. Cell Mol Gastroenterol Hepatol. 2020;10:391-418.doi;10.1016/j.jcmgh.2020.04.002

[50] Srinivasan AR, Tran TT, Bonini NM. Loss of miR-34 in Drosophila dysregulates protein translation and protein turnover in the aging brain. Aging Cell. 2022;21:e13559.doi;10.1111/acel.13559

[51] Cornejo PJ, Vergoni B, Ohanna M, Angot B, Gonzalez T, Jager J, Tanti JF, Cormont M. The Stress-Responsive miRNA-34a Alters Insulin Signaling and Actions in Adipocytes through Induction of the Tyrosine Phosphatase PTP1B. Cells. 2022; 11(16):2581. doi;10.3390/cells11162581

[52] Cui H, Ge J, Xie N, Banerjee S, Zhou Y, Liu RM, Thannickal VJ, Liu G. miR-34a promotes fibrosis in aged lungs by inducing alveolarepithelial dysfunctions. Am J Physiol Lung Cell Mol Physiol. 2017;312:L415-l424.doi;10.1152/ajplung.00335.2016

[53] Cui H, Ge J, Xie N, Banerjee S, Zhou Y, Antony VB, Thannickal VJ, Liu G. miR-34a Inhibits Lung Fibrosis by Inducing Lung Fibroblast Senescence. Am J Respir Cell Mol Biol. 2017;56:168-178.doi;10.1165/rcmb.2016-0163OC

[54] Iwagami Y, Zou J, Zhang H, Cao K, Ji C, Kim M, Huang CK. Alcohol-mediated miR-34a modulates hepatocyte growth and apoptosis. J Cell Mol Med. 2018;22:3987-3995.doi;10.1111/jcmm.13681

Journal Pre-proof

Journal Pre-proof

[55] Song L, Chen TY, Zhao XJ, Xu Q, Jiao RQ, Li JM, Kong LD. Pterostilbene prevents hepatocyte epithelial-mesenchymal transition in fructose-induced liver fibrosis through suppressing miR-34a/Sirt1/p53 and TGF-β1/Smads signalling. Br J Pharmacol. 2019;176:1619-1634.doi;10.1111/bph.14573

[56] Wan Y, Zhou T, Slevin E, Koyama S, Li X, Harrison K, Li T, Zhou B, Lorenzo SR, Zhang Y, Xu W, Klaunig JE, Wu C, Shetty AK, Huang CK, Meng F. Liver-specific deletion of miRNA-34a alleviates ductular reaction and liver fibrosis during experimental cholestasis. Faseb j. 2023;37:e22731.doi;10.1096/fj.202201453R

[57] Wan Y, Slevin E, Koyama S, Huang CK, Shetty AK, Li X, Harrison K, Li T, Zhou B, Lorenzo SR, Zhang Y, Salinas JM, Xu W, Klaunig JE, Wu C, Tsukamoto H, Meng F. miR-34a regulates macrophage-associated inflammation and angiogenesis in alcohol-induced liver injury. Hepatol Commun. 2023; 7(4):e0089. doi;10.1097/hc9.0000000000000089

[58] Setten E, Castagna A, Nava-Sedeño JM, Weber J, Carriero R, Reppas A, Volk V, Schmitz J, Gwinner W, Hatzikirou H, Feuerhake F, Locati M. Understanding fibrosis pathogenesis via modeling macrophage-fibroblast interplay in immune-metabolic context. Nat Commun. 2022;13:6499.doi;10.1038/s41467-022-34241-5

[59] Yan G, Li B, Xin X, Xu M, Ji G, Yu H. miRNA-34a Promotes Hepatic Stellate Cell Activation via Targeting ACSL1. Med Sci Monit. 2015;21:3008-3015.doi;10.12659/msm.894000

[60] Li X, Zhang W, Xu K, Lu J. miR-34a promotes liver fibrosis in patients with chronic hepatitis via mediating Sirt1/p53 signaling pathway. Pathol Res Pract. 2020;216:152876.doi;10.1016/j.prp.2020.152876

[61] Jiang XP, Ai WB, Wan LY, Zhang YQ, Wu JF. The roles of miRNA families in hepatic fibrosis. Cell Biosci. 2017;7:34.doi;10.1186/s13578-017-0161-7

[62] Song J, Lv H, Liu B, Hao M, Taylor HS, Zhang X, Li D, Huang Y. Let-7 suppresses liver fibrosis by inhibiting hepatocyte apoptosis and TGF-β production. Mol Metab. 2023;78:101828.doi;10.1016/j.molmet.2023.101828

[63] McDaniel K, Huang L, Sato K, Wu N, Annable T, Zhou T, Ramos-Lorenzo S, Wan Y, Huang Q, Francis H, Glaser S, Tsukamoto H, Alpini G, Meng F. The let-7/Lin28 axis regulates activation of hepatic stellate cells in alcoholic liver injury. J Biol Chem. 2017;292:11336-11347.doi;10.1074/jbc.M116.773291

[64] Matsuura K, De Giorgi V, Schechterly C, Wang RY, Farci P, Tanaka Y, Alter HJ. Circulating let-7 levels in plasma and extracellular vesicles correlate with hepatic fibrosis progression in chronic hepatitis C. Hepatology. 2016;64:732-745.doi;10.1002/hep.28660

[65] Sun J, Shi L, Xiao T, Xue J, Li J, Wang P, Wu L, Dai X, Ni X, Liu Q. miRNA-21, via the HIF-1α/VEGF signaling pathway, is involved in arsenite-induced hepatic fibrosis

Journal Pre-proof

Journal Pre-proof

through aberrant cross-talk of hepatocytes and hepatic stellate cells. Chemosphere. 2021;266:129177.doi;10.1016/j.chemosphere.2020.129177

[66] Xue J, Xiao T, Wei S, Sun J, Zou Z, Shi M, Sun Q, Dai X, Wu L, Li J, Xia H, Tang H, Zhang A, Liu Q. miR-21-regulated M2 polarization of macrophage is involved in arsenicosis-induced hepatic fibrosis through the activation of hepatic stellate cells. J Cell Physiol. 2021;236:6025-6041.doi;10.1002/jcp.30288

[67] He X, Xie J, Zhang D, Su Q, Sai X, Bai R, Chen C, Luo X, Gao G, Pan W. Recombinant adeno-associated virus-mediated inhibition of miRNA-21 protects mice against the lethal schistosome infection by repressing both IL-13 and transforming growth factor beta 1 pathways. Hepatology. 2015;61:2008-2017.doi;10.1002/hep.27671

[68] Bao S, Zheng J, Li N, Huang C, Chen M, Cheng Q, Yu K, Chen S, Zhu M, Shi G. Serum miRNA Levels as a Noninvasive Diagnostic Biomarker for the Early Diagnosis of Hepatitis B Virus-Related Liver Fibrosis. Gut Liver. 2017;11:860-869.doi;10.5009/gnl16560

[69] Dong Z, Li S, Wang X, Si L, Ma R, Bao L, Bo A. lncRNA GAS5 restrains CCl(4)-induced hepatic fibrosis by targeting miR-23a through the PTEN/PI3K/Akt signaling pathway. Am J Physiol Gastrointest Liver Physiol. 2019;316:G539-g550.doi;10.1152/ajpgi.00249.2018

[70] Li H, Li X, Yu S, Hu Y, Xu L, Wang T, Yang X, Sun X, Zhao B. miR-23b Ameliorates nonalcoholic steatohepatitis by targeting Acyl-CoA thioesterases 4. Exp Cell Res. 2021;407:112787.doi;10.1016/j.yexcr.2021.112787

[71] El-Hefny M, Fouad S, Hussein T, Abdel-Hameed R, Effat H, Mohamed H, Abdel Wahab AH. Circulating miRNAs as predictive biomarkers for liver disease progression of chronic hepatitis C (genotype-4) Egyptian patients. J Med Virol. 2019;91:93-101.doi;10.1002/jmv.25294

[72] Wang S, Li M, Zhao X, Wang H, Zhu J, Wang C, Zhou M, Dong H, Zhou R. Upregulation of KSRP by miR-27b attenuates schistosomiasis-induced hepatic fibrosis by targeting TGF-β1. Faseb j. 2020;34:4120-4133.doi;10.1096/fj.201902438R

[73] Hall C, Ehrlich L, Meng F, Invernizzi P, Bernuzzi F, Lairmore TC, Alpini G, Glaser S. Inhibition of miRNA-24 increases liver fibrosis by enhanced menin expression in Mdr2(-/-) mice. J Surg Res. 2017;217:160-169.doi;10.1016/j.jss.2017.05.020

[74] Wan LY, Peng H, Ni YR, Jiang XP, Wang JJ, Zhang YQ, Ma L, Li R, Han L, Tan Y, Li JM, Cai WL, Yuan WF, Liang JJ, Huang L, Wu X, Zhou Q, Cheng QN, Yang X, Liu MY, Ai WB, Liu CB, Zhang H, Wu JF. The miR-23b/27b/24-1 Cluster Inhibits Hepatic Fibrosis by Inactivating Hepatic Stellate Cells. Cell Mol Gastroenterol Hepatol. 2022;13:1393-1412.doi;10.1016/j.jcmgh.2022.01.016

[75] Kalvari I, Nawrocki EP, Ontiveros-Palacios N, Argasinska J, Lamkiewicz K, Marz M, Griffiths-Jones S, Toffano-Nioche C, Gautheret D, Weinberg Z, Rivas E, Eddy SR, Finn

Journal Pre-proof

Journal Pre-proof

RD, Bateman A, Petrov AI. Rfam 14: expanded coverage of metagenomic, viral and miRNA families. Nucleic Acids Res. 2021;49:D192-d200.doi;10.1093/nar/gkaa1047

[76] Reinhart BJ, Slack FJ, Basson M, Pasquinelli AE, Bettinger JC, Rougvie AE, Horvitz HR, Ruvkun G. The 21-nucleotide let-7 RNA regulates developmental timing in Caenorhabditis elegans. Nature. 2000;403:901-906.doi;10.1038/35002607

[77] Roush S, Slack FJ. The let-7 family of miRNAs. Trends Cell Biol. 2008;18:505-516.doi;10.1016/j.tcb.2008.07.007

[78] Chirshev E, Oberg KC, Ioffe YJ, Unternaehrer JJ. Let-7 as biomarker, prognostic indicator, and therapy for precision medicine in cancer. Clin Transl Med. 2019;8:24.doi;10.1186/s40169-019-0240-y

[79] Zhong H, Dong B, Zhu D, Fu Z, Liu J, Jin Y. Sja-let-7 suppresses the development of liver fibrosis via Schistosoma japonicum extracellular vesicles. PLoS Pathog. 2024;20:e1012153.doi;10.1371/journal.ppat.1012153

[80] Sarkies P, Miska EA. Molecular biology. Is there social RNA? Science. 2013;341:467-468.doi;10.1126/science.1243175

[81] Cai P, Mu Y, Olveda RM, Ross AG, Olveda DU, McManus DP. Circulating miRNAs as footprints for liver fibrosis grading in schistosomiasis. EBioMedicine. 2018;37:334-343.doi;10.1016/j.ebiom.2018.10.048

[82] Lagos-Quintana M, Rauhut R, Lendeckel W, Tuschl T. Identification of novel genes coding for small expressed RNAs. Science. 2001;294:853-858.doi;10.1126/science.1064921

[83] Liu J, Zhang S, Huang Y, Sun L. miR-21 protects neonatal rats from hypoxic-ischemic brain damage by targeting CCL3. Apoptosis. 2020;25:275-289.doi;10.1007/s10495-020-01596-3

[84] Ramanujam D, Schön AP, Beck C, Vaccarello P, Felician G, Dueck A, Esfandyari D, Meister G, Meitinger T, Schulz C, Engelhardt S. miRNA-21-Dependent Macrophage-to-Fibroblast Signaling Determines the Cardiac Response to Pressure Overload. Circulation. 2021;143:1513-1525.doi;10.1161/circulationaha.120.050682

[85] Jenike AE, Halushka MK. miR-21: a non-specific biomarker of all maladies. Biomark Res. 2021;9:18.doi;10.1186/s40364-021-00272-1

[86] Kumar D, Narang R, Sreenivas V, Rastogi V, Bhatia J, Saluja D, Srivastava K. Circulatory miR-133b and miR-21 as Novel Biomarkers in Early Prediction and Diagnosis of Coronary Artery Disease. Genes (Basel). 2020; 11(2):164. doi;10.3390/genes11020164

[87] Aalami AH, Abdeahad H, Mesgari M. Circulating miR-21 as a potential biomarker in human digestive system carcinoma: a systematic review and diagnostic meta-analysis.

Journal Pre-proof

Journal Pre-proof

Biomarkers. 2021;26:103-113.doi;10.1080/1354750x.2021.1875504

[88] Dooley S, Hamzavi J, Breitkopf K, Wiercinska E, Said HM, Lorenzen J, Ten Dijke P, Gressner AM. Smad7 prevents activation of hepatic stellate cells and liver fibrosis in rats. Gastroenterology. 2003;125:178-191.doi;10.1016/s0016-5085(03)00666-8

[89] Zhou X, Xiong J, Lu S, Luo L, Chen ZL, Yang F, Jin F, Wang Y, Ma Q, Luo YY, Wang YJ, Zhou JB, Liu P, Zhao L. Inhibitory Effect of Corilagin on miR-21-Regulated Hepatic Fibrosis Signaling Pathway. Am J Chin Med. 2019;47:1541-1569.doi;10.1142/s0192415x19500794

[90] Migita K, Komori A, Kozuru H, Jiuchi Y, Nakamura M, Yasunami M, Furukawa H, Abiru S, Yamasaki K, Nagaoka S, Hashimoto S, Bekki S, Kamitsukasa H, Nakamura Y, Ohta H, Shimada M, Takahashi H, Mita E, Hijioka T, Yamashita H, Kouno H, Nakamuta M, Ario K, Muro T, Sakai H, Sugi K, Nishimura H, Yoshizawa K, Sato T, Naganuma A, Komatsu T, Oohara Y, Makita F, Tomizawa M, Yatsuhashi H. Circulating miRNA Profiles in Patients with Type-1 Autoimmune Hepatitis. PLoS One. 2015;10:e0136908.doi;10.1371/journal.pone.0136908

[91] Mendes ND, Freitas AT, Sagot MF. Current tools for the identification of miRNA genes and their targets. Nucleic Acids Res. 2009;37:2419-2433.doi;10.1093/nar/gkp145

[92] Yoshida K, Yokoi A, Yamamoto Y, Kajiyama H. ChrXq27.3 miRNA cluster functions in cancer development. J Exp Clin Cancer Res. 2021;40:112.doi;10.1186/s13046-021-01910-0

[93] Ru L, Wang XM, Niu JQ. The miR-23-27-24 cluster: an emerging target in NAFLD pathogenesis. Acta Pharmacol Sin. 2022;43:1167-1179.doi;10.1038/s41401-021-00819-w

[94] Chhabra R, Dubey R, Saini N. Cooperative and individualistic functions of the miRNAs in the miR-23a~27a~24-2 cluster and its implication in human diseases. Mol Cancer. 2010;9:232.doi;10.1186/1476-4598-9-232

[95] Cui M, Yao X, Lin Y, Zhang D, Cui R, Zhang X. Interactive functions of miRNAs in the miR-23a-27a-24-2 cluster and the potential for targeted therapy in cancer. J Cell Physiol. 2020;235:6-16.doi;10.1002/jcp.28958

[96] Bustin SA, Benes V, Garson JA, et al. The MIQE guidelines: minimum information for publication of quantitative real-time PCR experiments. Clin Chem. 2009;55:611-622. doi:10.1373/clinchem.2008.112797

[97] Schmittgen TD, Jiang J, Liu Q, Yang L. A high-throughput method to monitor the expression of miRNA precursors. Nucleic Acids Res. 2004;32:e43.doi;10.1093/nar/gnh040

[98] He F, Ni N, Wang H, Zeng Z, Zhao P, Shi D, Xia Y, Chen C, Hu DA, Qin KH, Wagstaff W, Qin D, Hendren-Santiago B, Ho SH, Haydon RC, Luu HH, Reid RR, Shen L, Gan

Journal Pre-proof

Journal Pre-proof

H, Fan J, He TC. OUHP: an optimized universal hairpin primer system for cost-effective and high-throughput RT-qPCR-based quantification of miRNA (miRNA) expression. Nucleic Acids Res. 2022;50:e22.doi;10.1093/nar/gkab1153

[99] Yin JQ, Zhao RC, Morris KV. Profiling miRNA expression with microarrays. Trends Biotechnol. 2008;26:70-76.doi;10.1016/j.tibtech.2007.11.007

[100] Jung S, Kim WJ, Kim BK, Kim J, Kim MJ, Kim KP, Kim SK. In-particle stem-loop RT-qPCR for specific and multiplex miRNA profiling. Biosens Bioelectron. 2020;163:112301.doi;10.1016/j.bios.2020.112301

[101] Sarkar D, Parkin R, Wyman S, Bendoraite A, Sather C, Delrow J, Godwin AK, Drescher C, Huber W, Gentleman R, Tewari M. Quality assessment and data analysis for miRNA expression arrays. Nucleic Acids Res. 2009;37:e17.doi;10.1093/nar/gkn932

[102] Larson NB, Oberg AL, Adjei AA, Wang L. A Clinician's Guide to Bioinformatics for Next-Generation Sequencing. J Thorac Oncol. 2023;18:143-157.doi;10.1016/j.jtho.2022.11.006

[103] Hindson BJ, Ness KD, Masquelier DA, et al. High-throughput droplet digital PCR system for absolute quantitation of DNA copy number. Anal Chem. 2011;83:8604-8610. doi:10.1021/ac202028g

[104] Pratt DS, Kaplan MM. Evaluation of abnormal liver-enzyme results in asymptomatic patients. N Engl J Med. 2000;342:1266-1271.doi;10.1056/nejm200004273421707

[105] Raza A, Khan AQ, Inchakalody VP, Mestiri S, Yoosuf Z, Bedhiafi T, El-Ella DMA, Taib N, Hydrose S, Akbar S, Fernandes Q, Al-Zaidan L, Krishnankutty R, Merhi M, Uddin S, Dermime S. Dynamic liquid biopsy components as predictive and prognostic biomarkers in colorectal cancer. J Exp Clin Cancer Res. 2022;41:99.doi;10.1186/s13046-022-02318-0

[106] Rath RJ, Herrington JO, Adeel M, Güder F, Dehghani F, Farajikhah S. Ammonia detection: A pathway towards potential point-of-care diagnostics. Biosens Bioelectron. 2024;251:116100.doi;10.1016/j.bios.2024.116100

[107] Zhou J, Yu L, Gao X, Hu J, Wang J, Dai Z, Wang JF, Zhang Z, Lu S, Huang X, Wang Z, Qiu S, Wang X, Yang G, Sun H, Tang Z, Wu Y, Zhu H, Fan J. Plasma miRNA panel to diagnose hepatitis B virus-related hepatocellular carcinoma. J Clin Oncol. 2011;29:4781-4788.doi;10.1200/jco.2011.38.2697

[108] Yang J, Hamade M, Wu Q, Wang Q, Axtell R, Giri S, Mao-Draayer Y. Current and Future Biomarkers in Multiple Sclerosis. Int J Mol Sci. 2022; 23(11):5877. doi;10.3390/ijms23115877

Journal Pre-proof

Journal Pre-proof

[109] Hussen BM, Hidayat HJ, Salihi A, Sabir DK, Taheri M, Ghafouri-Fard S. miRNA: A signature for cancer progression. Biomed Pharmacother. 2021;138:111528.doi;10.1016/j.biopha.2021.111528

[110] Sterling RK, Lissen E, Clumeck N, et al. Development of a simple noninvasive index to predict significant fibrosis in patients with HIV/HCV coinfection. Hepatology. 2006;43:1317-1325. doi:10.1002/hep.21178

[111] Wai CT, Greenson JK, Fontana RJ, et al. A simple noninvasive index can predict both significant fibrosis and cirrhosis in patients with chronic hepatitis C. Hepatology. 2003;38:518-526. doi:10.1053/jhep.2003.50346

[112] Teng KY, Ghoshal K. Role of Noncoding RNAs as Biomarker and Therapeutic Targets for Liver Fibrosis. Gene Expr. 2015;16:155-162. doi:10.3727/105221615X14399878166078

[113] Musaddaq G, Shahzad N, Ashraf MA, Arshad MI. Circulating liver-specific microRNAs as noninvasive diagnostic biomarkers of hepatic diseases in human. Biomarkers. 2019;24:103-109. doi:10.1080/1354750X.2018.1528631

[114] Laterza OF, Lim L, Garrett-Engele PW, Vlasakova K, Muniappa N, Tanaka WK, Johnson JM, Sina JF, Fare TL, Sistare FD, Glaab WE. Plasma miRNAs as sensitive and specific biomarkers of tissue injury. Clin Chem. 2009;55:1977-1983.doi;10.1373/clinchem.2009.131797

[115] Jarry J, Schadendorf D, Greenwood C, Spatz A, van Kempen LC. The validity of circulating miRNAs in oncology: five years of challenges and contradictions. Mol Oncol. 2014;8:819-829.doi;10.1016/j.molonc.2014.02.009

[116] Kramvis A, Chang KM, Dandri M, Farci P, Glebe D, Hu J, Janssen HLA, Lau DTY, Penicaud C, Pollicino T, Testoni B, Van Bömmel F, Andrisani O, Beumont-Mauviel M, Block TM, Chan HLY, Cloherty GA, Delaney WE, Geretti AM, Gehring A, Jackson K, Lenz O, Maini MK, Miller V, Protzer U, Yang JC, Yuen MF, Zoulim F, Revill PA. A roadmap for serum biomarkers for hepatitis B virus: current status and future outlook. Nat Rev Gastroenterol Hepatol. 2022;19:727-745.doi;10.1038/s41575-022-00649-z

Figure 1. Current approaches to liver-fibrosis assessment. Abbreviations: miRNA, microRNA; HSC, hepatic stellate cell.

Figure 2. miRNA biogenesis. Abbreviations: pri-miRNA, primary miRNA; RISC, RNA-induced silencing complex.

Figure 3. Opposing miRNA effects on liver fibrosis. Abbreviations: HSC, hepatic stellate cell; TGF-beta, transforming growth factor beta.

Figure 4. Reported serum or plasma changes in selected miRNAs during liver disease. Upward and downward arrows indicate relative increases and decreases in the cited cohorts.

Figure 5. Platforms for circulating miRNA detection. Abbreviations: RT-qPCR, reverse-transcription quantitative PCR; NGS, next-generation sequencing; ddPCR, droplet digital PCR; CRISPR, clustered regularly interspaced short palindromic repeats.

Table 1. Selected miRNAs in hepatic fibrogenesis and circulating biomarker studies

miRNA	Mechanistic findings	Circulating findings
miR-122	Protects against ethanol-induced liver injury by suppressing HIF-1α [31]. Regulates hepatic glutamine metabolism [33]. Targets P4HA1 and reduces collagen production in HSCs [34].	Serum levels were higher in CHC than in controls, but did not significantly distinguish mild from moderate fibrosis [35].
miR-29a	Suppresses CD36 and attenuates diet-induced steatohepatitis and liver fibrosis in mice [42].	Serum miR-29a was increased in CHC, whereas hepatic miR-29a was reduced, consistent with enhanced vesicular release from activated HSCs [46].
miR-34a	Promotes hepatocyte senescence and apoptosis [54]. Regulates macrophage polarization in alcohol-associated liver injury [57]. Targets ACSL1 and promotes HSC activation and collagen expression [59,61].	Serum miR-34a was increased in CHC and associated with fibrosis-related indices [60].

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miRNA	Mechanistic findings	Circulating findings
let-7 family	Inhibits hepatocyte apoptosis and TGF-β1 production [62]. *let-7a*/*let-7b* suppress myofibroblastic HSC activation [63].	Plasma *let-7a*/*let-7c*/*let-7d-5p* levels were inversely associated with fibrosis stage and declined during fibrosis progression in CHC [64].
miR-21	Promotes HSC activation via HIF-1α/VEGF signaling [65] and profibrotic M2 macrophage polarization [66] during arsenite exposure. Promotes profibrotic signaling through SMAD7 suppression in schistosomiasis [67].	Serum miR-21 was lower in S3–S4 than in S0–S2 fibrosis in chronic HBV infection [68]. The direction of change depends on disease context.
miR-23a	Suppresses PTEN and promotes HSC activation, with increased α-SMA and collagen I expression [69].	—
miR-23b	Regulates acyl-CoA metabolism through ACOT4 and attenuates steatohepatitis and fibrosis in mice [70].	Serum miR-23b was inversely correlated with AST in CHC [71].
miR-27b	miR-27b knockdown or KSRP overexpression attenuated schistosomiasis-induced hepatic fibrosis [72].	—
miR-27a	—	Serum miR-27a was inversely associated with fibrosis and steatosis stages in CHC [71].
miR-24	Inhibition increases menin and TGF-β1 expression and worsens biliary fibrosis in Mdr2-deficient mice [73].	—

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miRNA	Mechanistic findings	Circulating findings
miR-23b/27b/24-1 cluster	Suppresses HSC activation and fibrosis by targeting TGF-β2, Gremlin1, LOX, integrin α2, and integrin α5 [74].	—

Notes: Mechanistic findings mainly derive from experimental studies; circulating findings are observational and do not establish a validated fibrosis-staging test. Tissue and circulating changes may differ. —, not assessed in the studies cited for that row.

Abbreviations: ACOT4, acyl-CoA thioesterase 4; ACSL1, acyl-CoA synthetase long-chain family member 1; AST, aspartate aminotransferase; CD36, cluster of differentiation 36; CHC, chronic hepatitis C; HBV, hepatitis B virus; HIF-1α, hypoxia-inducible factor 1α; HSC, hepatic stellate cell; KSRP, KH-type splicing regulatory protein; LOX, lysyl oxidase; P4HA1, prolyl 4-hydroxylase subunit α1; PTEN, phosphatase and tensin homolog; α-SMA, α-smooth muscle actin; TGF-β, transforming growth factor β; VEGF, vascular endothelial growth factor.

Table 2. Comparison of established platforms for miRNA measurement

Method	Principle and main use	Strengths	Limitations
RT-qPCR	Reverse transcription followed by real-time PCR; targeted measurement of known miRNAs [96,98].	Sensitive targeted assays; broad reagent availability; practical for candidate validation.	Requires validated primers, amplification efficiency, controls, and normalization. Reverse-transcription bias, inhibitors, and cross-reactivity can affect results. Cannot discover unknown sequences.
Microarray	Hybridization to predefined probes; simultaneous relative expression profiling of known miRNAs [99,101].	Profiles many known miRNAs in parallel; useful for comparative screening.	Probe-dependent coverage; cross-hybridization, background signal, and limited dynamic range. Cannot discover unknown miRNA sequences; candidate findings require orthogonal validation.

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Method	Principle and main use	Strengths	Limitations
Small-RNA sequencing (NGS)	Sequencing of small-RNA libraries; broad expression profiling and discovery of candidate novel miRNAs and sequence variants [17].	High-throughput profiling with sequence information; can distinguish isomiRs and identify unannotated candidates.	Library preparation and sequence-dependent bias, depth requirements, bioinformatics, and cost. Relative read counts require normalization; novel candidates need annotation and experimental validation.

Notes: Platform capability does not establish the clinical validity of a liver-fibrosis biomarker. Each assay requires matrix-specific analytical validation and independent clinical validation. Emerging digital PCR, CRISPR-based, and isothermal approaches are discussed in the main text.

Abbreviations: CRISPR, clustered regularly interspaced short palindromic repeats; isomiRs, miRNA sequence variants; miRNA, microRNA; NGS, next-generation sequencing; PCR, polymerase chain reaction; RT-qPCR, reverse-transcription quantitative PCR.

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